Required practical: Preparing a pure, dry sample of a salt from an insoluble oxide or carbonate

C4 Chemical Changes

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Title - Making Salt Required Practical Write Up

Questions	Answers
1. Explain why an excess of the solid is added to the acid.	If the solid is in excess, all of the acid will have reacted and totally converted into the soluble salt. Also, if some acid was unreacted, it would contaminate the salt solution. The excess solid is easy to filter off.
2. Explain why the solution that has been filtered needs to be concentrated by heating a little.	It is heated to make the salt solution more concentrated. This means that the crystals form more effectively (saturated solution).

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Title - Making Salt Required Practical Write Up

Put the following steps for making salt in the correct order:

A. The concentrate is left to cool and crystallise to produce a solid salt.

B. The excess solid is filtered off leaving a solution of the salt.

C. The salt solution is then heated a little to concentrate it.

D. The solid is added to the acid until no more of the solid reacts and it is in excess.

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Which salts can be made?

Copy and complete the equations below:

Metal + acid → _________ + Hydrogen

Metal oxide + acid → _________ + water

Metal hydroxide + _________ → metal salt + water

Metal carbonate + acid → metal salt + _________ + _________

Forming ions:

Metals:		Acids:		
Magnesium	Mg^{2+}	Sulfuric acid	H_2SO_4	SO_4^{2-}
Iron	Fe^{2+}	Hydrochloric acid	HCl	Cl^-
Zinc	Zn^{2+}	Nitric acid	HNO_3	NO_3^-

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Forming ions:

- When determining the compound formed, the charges of the cation and anion must be equal
- Mg^{2+} and SO_4^{2-} have opposite and equal charges and so form $MgSO_4$

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I do

Forming ions:

- What will happen with Mg^{2+} and Cl^-?

We do

Title – Neutralisation

Forming ions:

- What will the chemical formula be with Zn^{2+} and NO_3^{-}?
- What will the chemical formula be with Na^{+} and Cl^{-}?
- What will the chemical formula be with Fe^{2+} and SO_4^{2+}?

Alkalis and bases

Metal oxides and metal hydroxides are **bases**.

Some bases are soluble in water, and these are called **alkalis**,

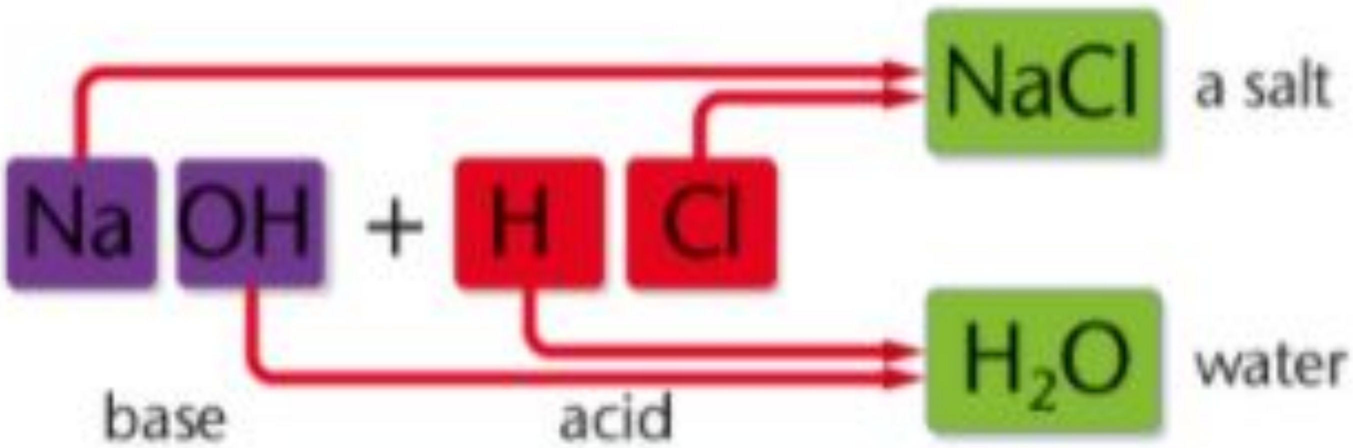
When an acid and a base react, *neutralisation* takes place to make a **salt** and water.

The word equation for neutralisation is:

acid + base(metal hydroxide) → salt + water

The reaction of sodium hydroxide, a soluble base, with hydrochloric acid produces sodium chloride.

sodium hydroxide + hydrochloric acid → sodium chloride + water



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We do

Write the word equation for the reaction between potassium hydroxide and hydrochloric acid.

acid + base → salt + water

Answer

Potassium hydroxide + Hydrochloric acid → Potassium chloride + water

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You do

1)**Write the word equation for the reaction between sodium hydroxide and hydrochloric acid.**

acid + base → salt + water

Sodium hydroxide + Hydrochloric acid → Sodium chloride + Water

Salts can also be made by reacting acids with excess of an insoluble base.

For example, zinc oxide is an insoluble base. Zinc oxide and sulfuric acid react to make zinc sulfate and water.

zinc oxide + sulfuric acid → zinc sulfate + water

zinc oxide + sulfuric acid → zinc sulfate + water

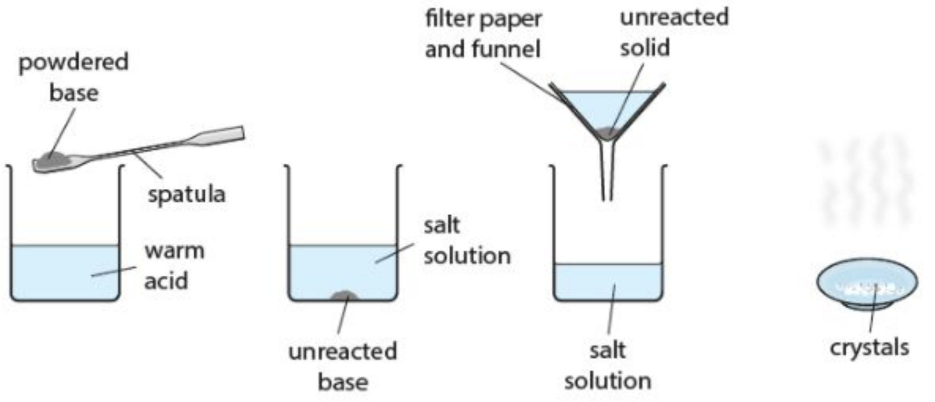

Figure 4.28 Making a salt with excess base or carbonate

Common chemical compounds

Acids		Bases and alkalis		Carbonates		Metals	
sulfuric acid	H_2SO_4	potassium hydroxide	KOH	sodium carbonate	Na_2CO_3	magnesium	Mg
nitric acid	HNO_3	sodium hydroxide	NaOH	calcium carbonate	$CaCO_3$	zinc	Zn
hydrochloric acid	HCl	calcium hydroxide	$Ca(OH)_2$	zinc carbonate	$ZnCO_3$	iron	Fe
		copper oxide	CuO	copper carbonate	$CuCO_3$		

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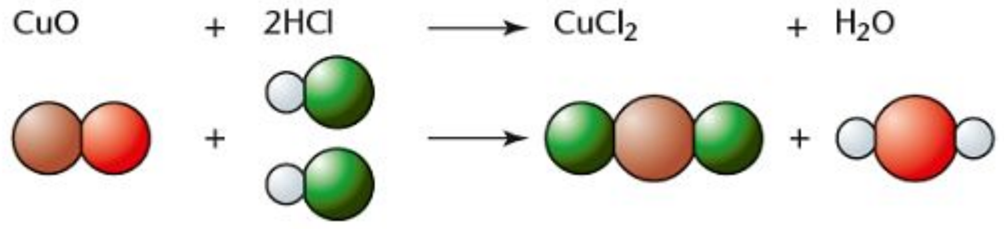
Charges on ions

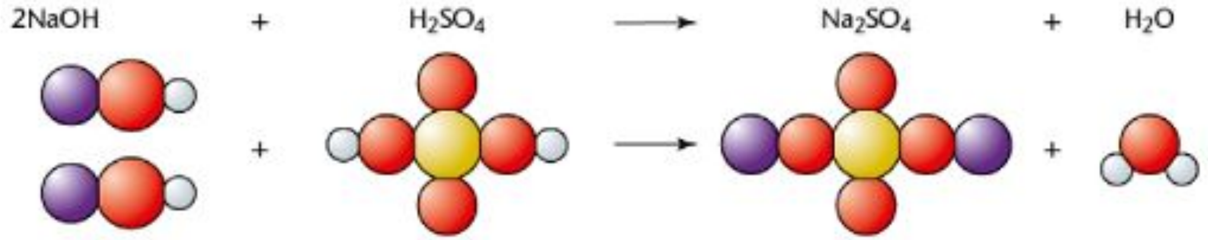
Salts are formed using ions with +2 and −1 charge (e.g. copper chloride CuCl2) or using ions with +1 and −2 charge (e.g. sodium sulfate Na2SO4), the equations will need to be balanced.

Single charge	Cl^-	NO_3^-	OH^-	Na^+	K^+	H^+
Double charge	$SO4^{2-}$	Ca^{2+}	Mg^{2+}	Zn^{2+}	Fe^{2+}	Cu^{2+}

$CuO + 2HCl \rightarrow CuCl_2 + H_2O$

$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + H_2O$





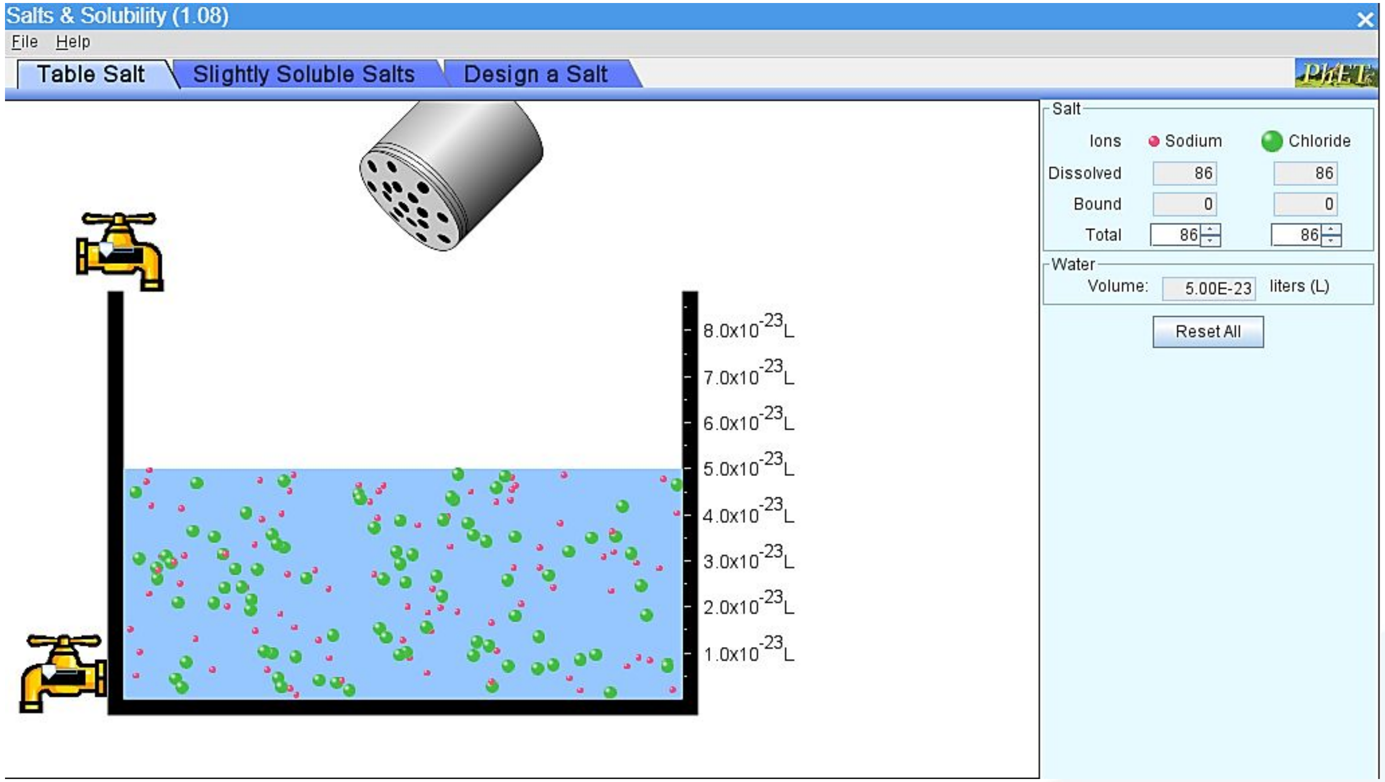
Charges on ions

Questions	Answers
1. Write a balanced equation for the reaction between zinc carbonate and hydrochloric acid.	$ZnCO_3 + 2HCl \rightarrow ZnCl_2 + H_2O + CO_2$
2. Write a balanced equation for the reaction between magnesium oxide and nitric acid.	$MgO + 2HNO_3 \rightarrow Mg(NO_3)_2 + H_2O$

Solubility simulation

Extended investigation of soluble salts:

https://phet.colorado.edu/sims/cheerpj/soluble-salts/latest/soluble-salts.html?simulation=soluble-salts



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Specification link

4.4.2.3 Soluble salts

Content	Key opportunities for skills development
Soluble salts can be made from acids by reacting them with solid insoluble substances, such as metals, metal oxides, hydroxides or carbonates. The solid is added to the acid until no more reacts and the excess solid is filtered off to produce a solution of the salt. Salt solutions can be crystallised to produce solid salts. Students should be able to describe how to make pure, dry samples of named soluble salts from information provided.	

Required practical 1: preparation of a pure, dry sample of a soluble salt from an insoluble oxide or carbonate using a Bunsen burner to heat dilute acid and a water bath or electric heater to evaporate the solution.

AT skills covered by this practical activity: 2, 3, 4 and 6.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in Key opportunities for skills development.

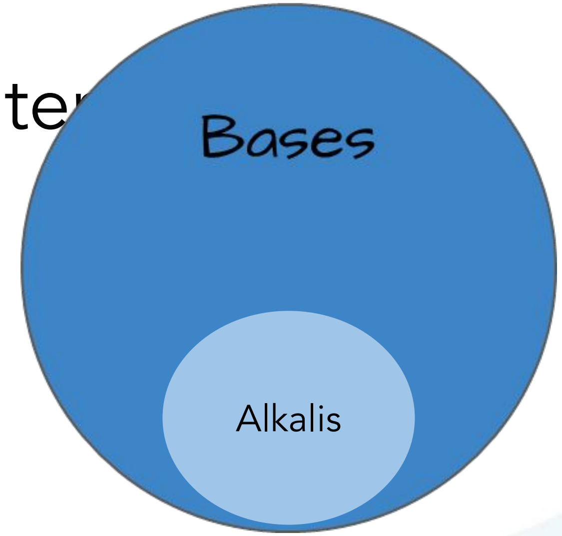
Bases and alkalis

- We can also use bases and alkalis to neutralise acids and produce salts. - you did it this morning when you brushed your teeth!

- Bases are ionic compounds that can neutralise acic

- Most bases are insoluble in wate

- Alkalis are soluble bases



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Making soluble salts

Soluble salts can be made by **reacting** acids with either **soluble** or **insoluble** **bases**.

Making a salt from an alkali

If you are using an alkali - which is a soluble base - then you need to add just enough acid to make a neutral solution (check a small sample with **universal indicator paper**).

Warm the salt solution to **evaporate** the water. You get larger crystals if you evaporate the water slowly.

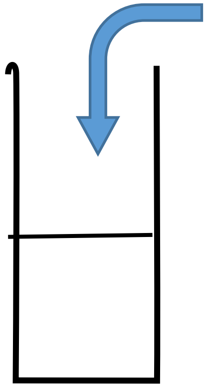
Making a salt from an insoluble metal oxide or carbonate

Copper oxide and other transition metal oxides or hydroxides do not dissolve in water. If the **base** is insoluble, then an extra step is needed to form a salt.

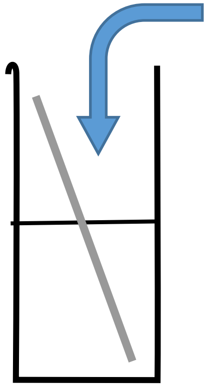
You add the base to the warm acid until no more will dissolve and you have some base left over – this is called an 'excess'. You filter the mixture to remove the excess base, and then evaporate the water in the **filtrate** to leave the salt behind.

Key questions

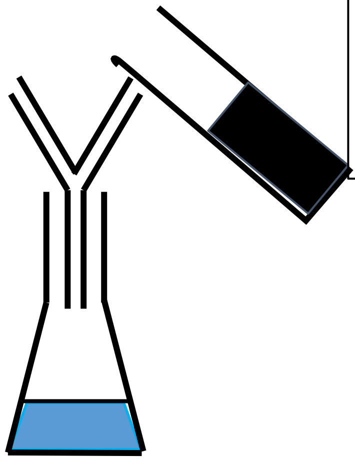
1. What is meant by the filtrate?
2. What is meant by the residue?
3. What is the filtrate in this experiment?
4. What is the residue in this experiment?
5. Why is a water bath used to evaporate the water from the copper sulphate solution instead of heating the evaporating basin directly with a Bunsen burner?
6. Why should you not evaporate all of the water from the copper sulphate solution?



1. Add 20cm^3 sulphuric acid to a beaker.
The acid is in a water bath - why?

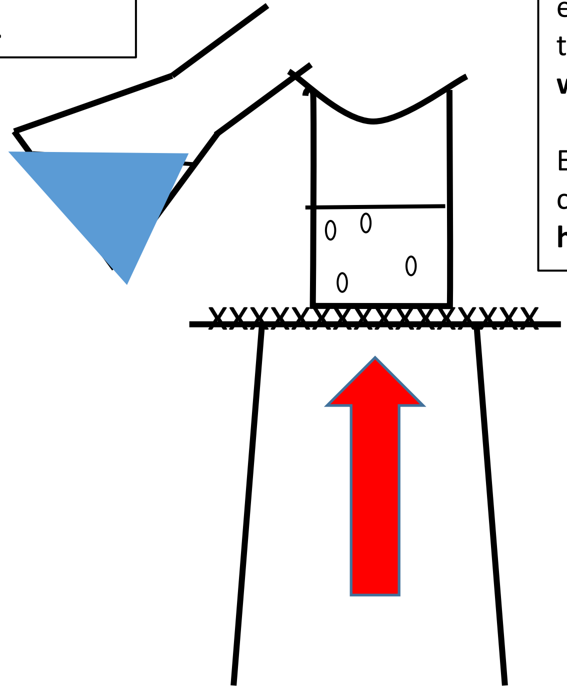


2. Add the copper oxide and stir until it is in **excess - how can you tell? Why do we need to do this?**



3. Filter the copper sulphate solution.

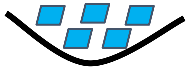
4. Pour the filtered solution into an evaporating dish.



5. Set up your equipment as shown in the diagram. Your evaporating dish needs to be on top of a half filled **beaker of water - why?**

Evaporate gently until the copper sulphate solution has **halved.**

6. Leave to crystallise.

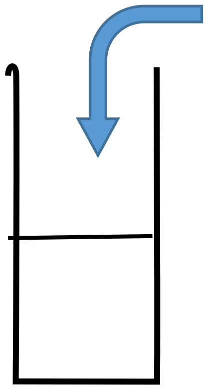


Questions

1. Write a balanced symbol equation for the reaction.

2. Why don't we see bubbles?

3. Write an ionic equation for the reaction.



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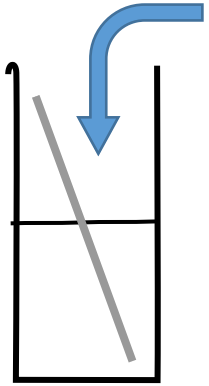

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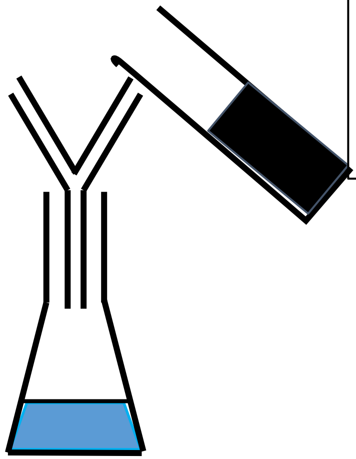
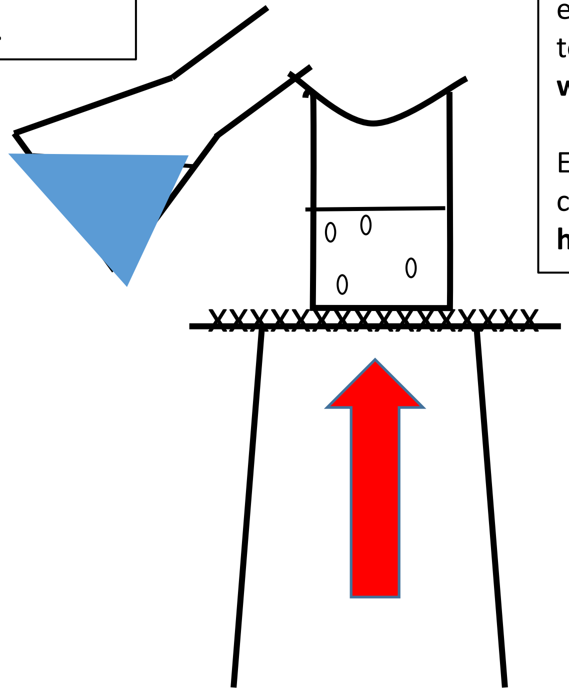


3. Filter the copper sulphate solution.

4. Pour the filtered solution into an evaporating dish.



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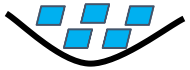

Questions

1. Write a balanced symbol equation for the reaction.
2. Why don't we see bubbles?
3. Write an ionic equation for the reaction.

Conclusion

1. What safety precautions should you take when carrying out this experiment and why?
2. Why was it necessary to warm the sulphuric acid?
3. What colour was the copper sulphate solution that formed?
4. Why was it necessary to add copper oxide until it was present in excess?
5. How did you know when the copper oxide was present in excess?
6. How did you separate the excess copper oxide from the copper sulphate solution?

Conclusion

1. What safety precautions should you take when carrying out this experiment and why?
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6. How did you separate the excess copper oxide from the copper sulphate solution?

Answers – self mark

1. Balanced symbol: $CuO_{(s)} + H_2SO_{4(aq)} \rightarrow CuSO_{4(aq)} + H_2O_{(l)}$
2. Neither product is a gas.
3. Ionic: $O^{2-} + 2H^+ \rightarrow H_2O_{(l)}$

Correct the mistakes

To prepare crystals of copper sulphate the method is as follows. Add nickel oxide to sulphuric hydroxide with gentle cooling until no more will dissolve. Distil the mixture to remove unreacted copper oxide. Pour the filtrate into an evacuating basin and heat the solution until crystals just start to disappear. Allow to warm to room temperature and leave to evaporate quickly to obtain red crystals of copper chloride.

CONSOLIDATION

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CONSOLIDATION

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CONSOLIDATION

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CONSOLIDATION

Exam question

(d) $\boxed{\text{Describe}}$ how crystals of calcium chloride can be made from calcium carbonate and dilute hydrochloric acid.

Always start by writing out the word equation first!

Self-mark in green pen

(d) Describe how crystals of calcium chloride can be made from calcium carbonate and dilute hydrochloric acid.

(d) add excess calcium carbonate to acid (and stir) / add $CaCO_3$ until fizzing stops

ignore heating the acid

accept answer using calcium oxide in place of calcium carbonate

1

(remove excess calcium carbonate by) filter(ing)

1

warm until a saturated solution forms / point of crystallisation / crystals start to form

*do **not** accept heat until all water gone*

1

leave to cool

dependent on previous mark

*If solution **not** heated allow leave to evaporate (1)*

until crystals form (1)

1